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PeriWatch Surveillance™

Interface Specification

PeriWatch Surveillance 2024.1.000

PeriGen
Advanced Perinatal Systems
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1 Single Sign On

Clinicians charting in their Electronic Health Record (EHR) system often find use for quickly viewing the maternal-fetal surveillance strip, as well as viewing or modifying annotations to this strip. To that end, PeriWatch Surveillance (PWS) has developed a single sign-on (SSO) integration whereby the user of an outside charting system may launch a special PWS surveillance screen without need of authenticating to PWS. This process and the enabling mechanism are described below.

1.2 Design Detail

Users of an EHR wishing to launch the PWS SSO integration will be provided a launch link, button, or other mechanism by the host EHR system. The host EHR system will launch an executable, the Message Dispatcher, in one of two modes:

1.2.3 Command Line Invocation using Message Dispatcher

One time use where the launch information is sent via command line. The Message Dispatcher then calls the Launch Service to pass the message along. In this case, there will be a launch per use of SSO launch. The command-line arguments contain the context and authentication required to successfully launch a relevant session of PWS.

1.2.4 File Watch Invocation

PeriWatch supports file-watch SSO mode, where the EHR can drop a file in a predesignated area indicating the user actions like Login, Logout and Patient Switch. These actions will be translated into appropriate PeriWatch launch messages and sent to the Launch Service. The files contain the context and authentication, in a predefined format, required to successfully launch a relevant session of PWS.

The Message Dispatcher resides on and is invoked from a central server and will communicate with a Launch Service on the PWS Server, The PWS server informs a resident and always-running Message Receiver on the user's client workstation. The Message Receiver already has registered with the PWS server using the value in the environment variable COMPUTERNAME.

The Message Receiver, upon receiving the encoded instructions, will launch PWS on the client workstation in a special windowed session, affect a login, and automatically display the surveillance strip of the patient indicated by the EHR at time of launch.

Command Line Invocation using PeriWatch Surveillance Client

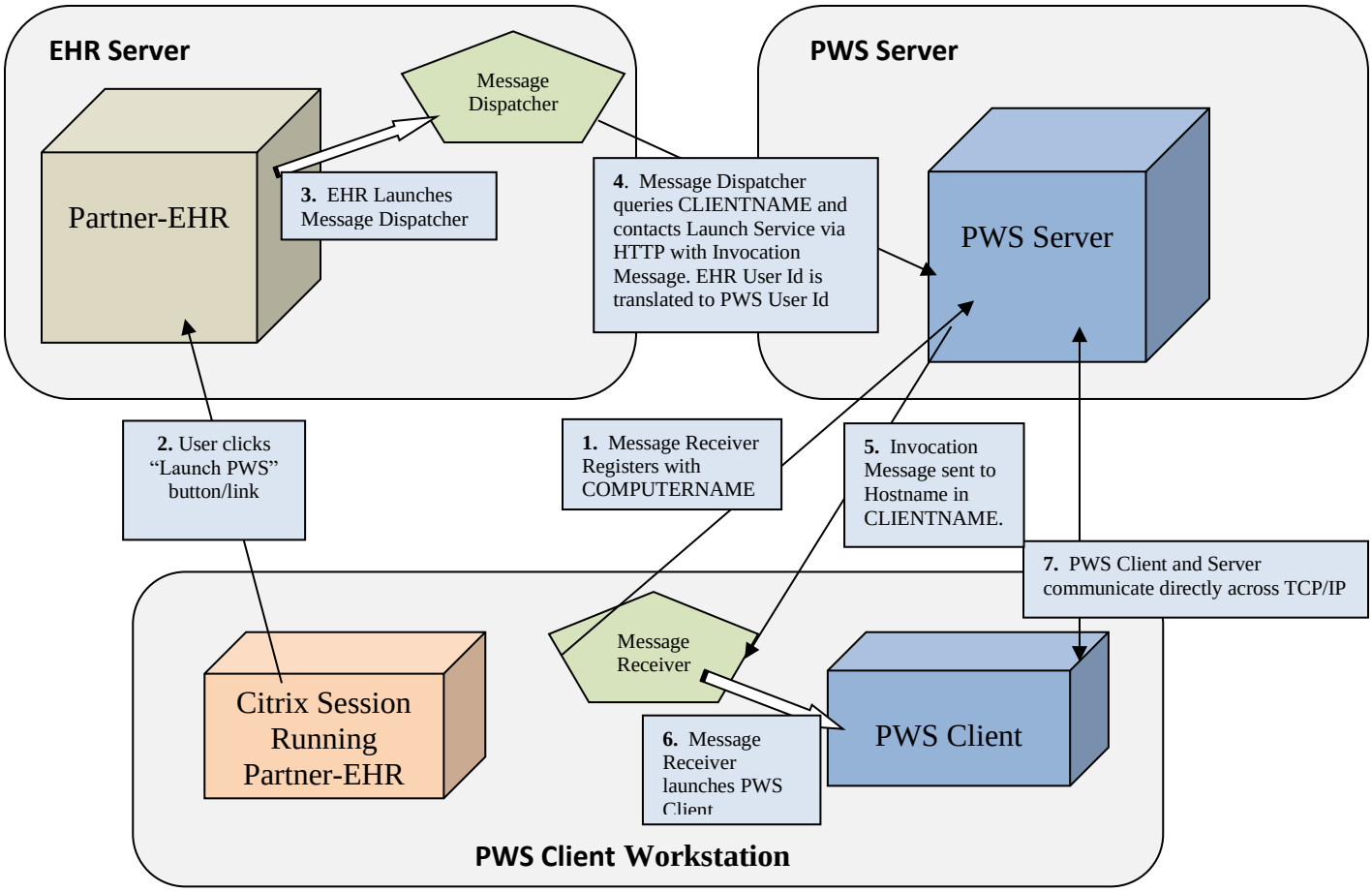
PeriWatch Surveillance application is launched using command line arguments from the EHR. The command-line arguments contain the context and authentication required to successfully launch a relevant session of PWS.

1.2.5 Web Invocation

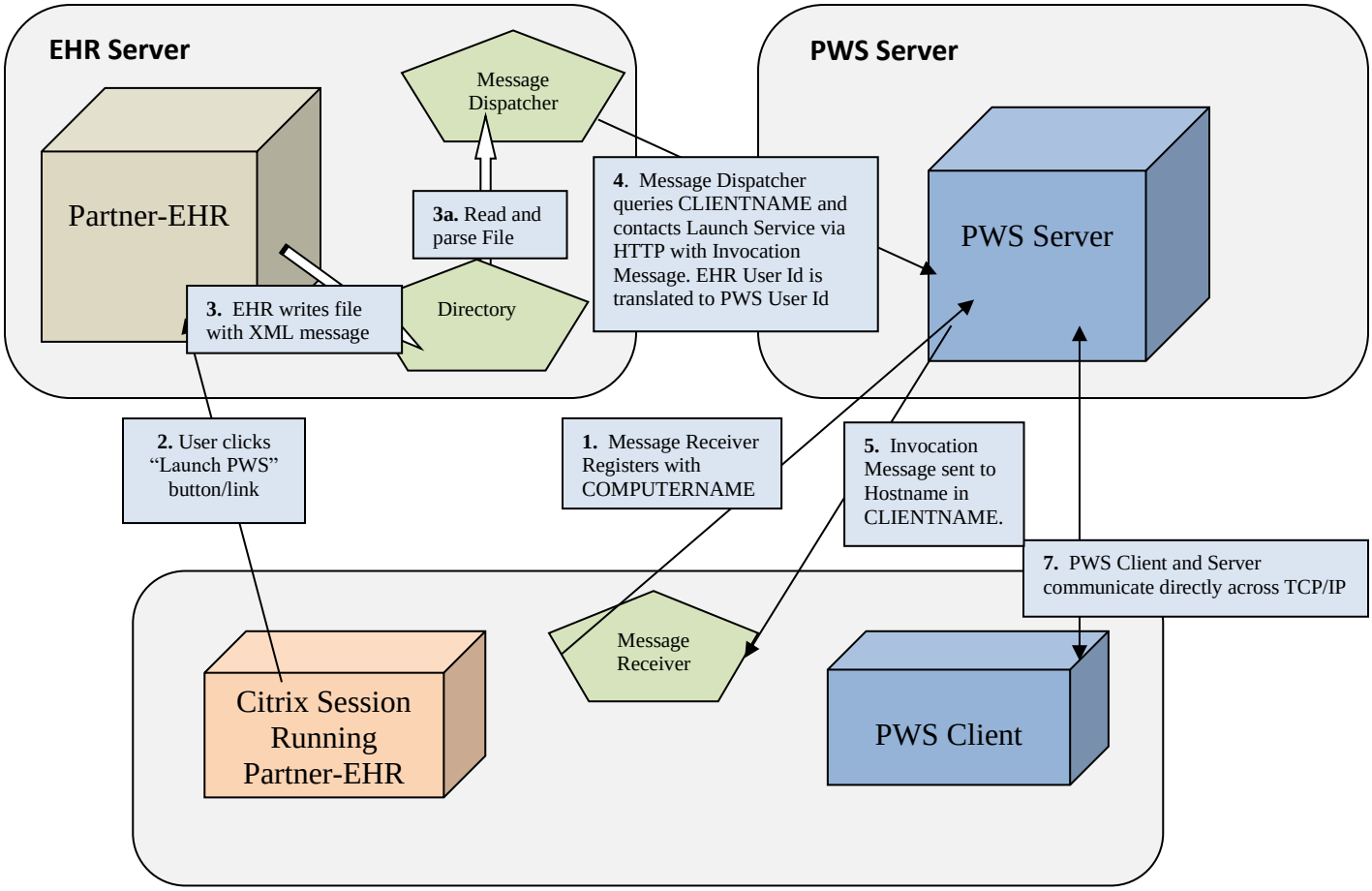
PeriWatch supports SSO launching from a web EHR using arguments sent in the URL. The arguments contain the context and authentication required to successfully launch a relevant session of PWS.

1.3 Invocation modes

1.3.1 Command Line Invocation using Message Dispatcher



1.3.2 File Watch Invocation



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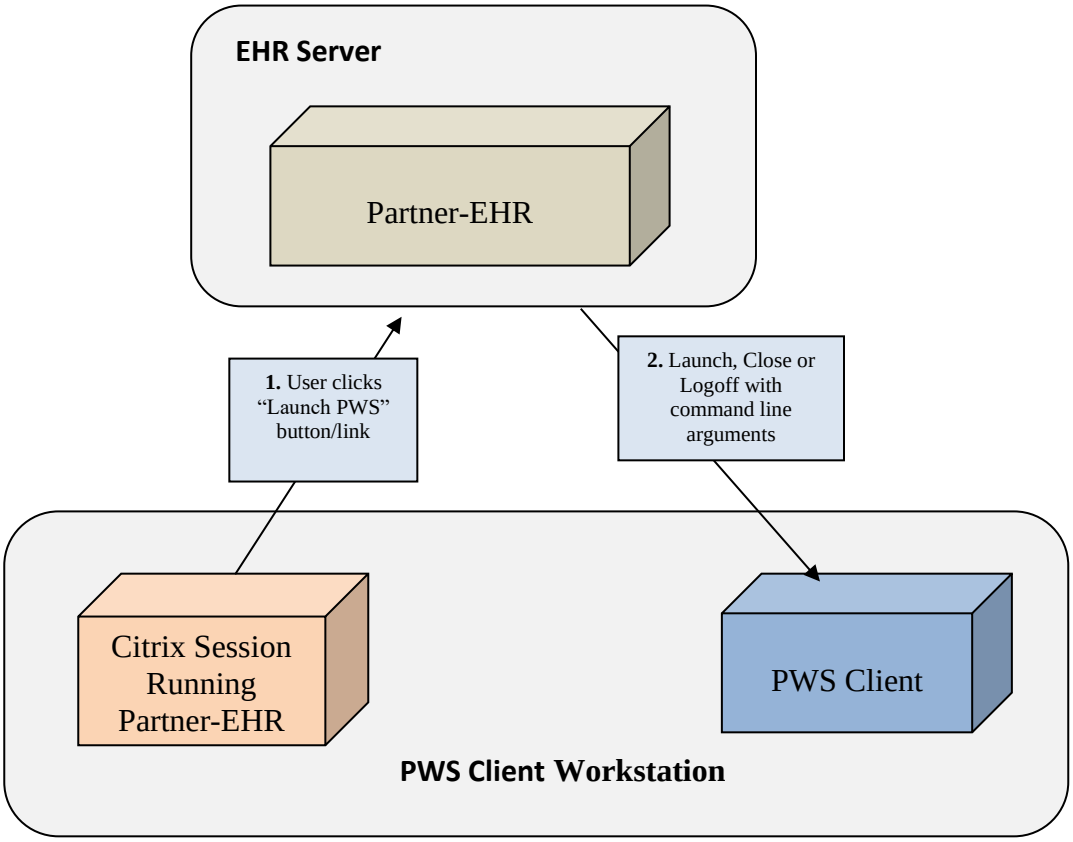
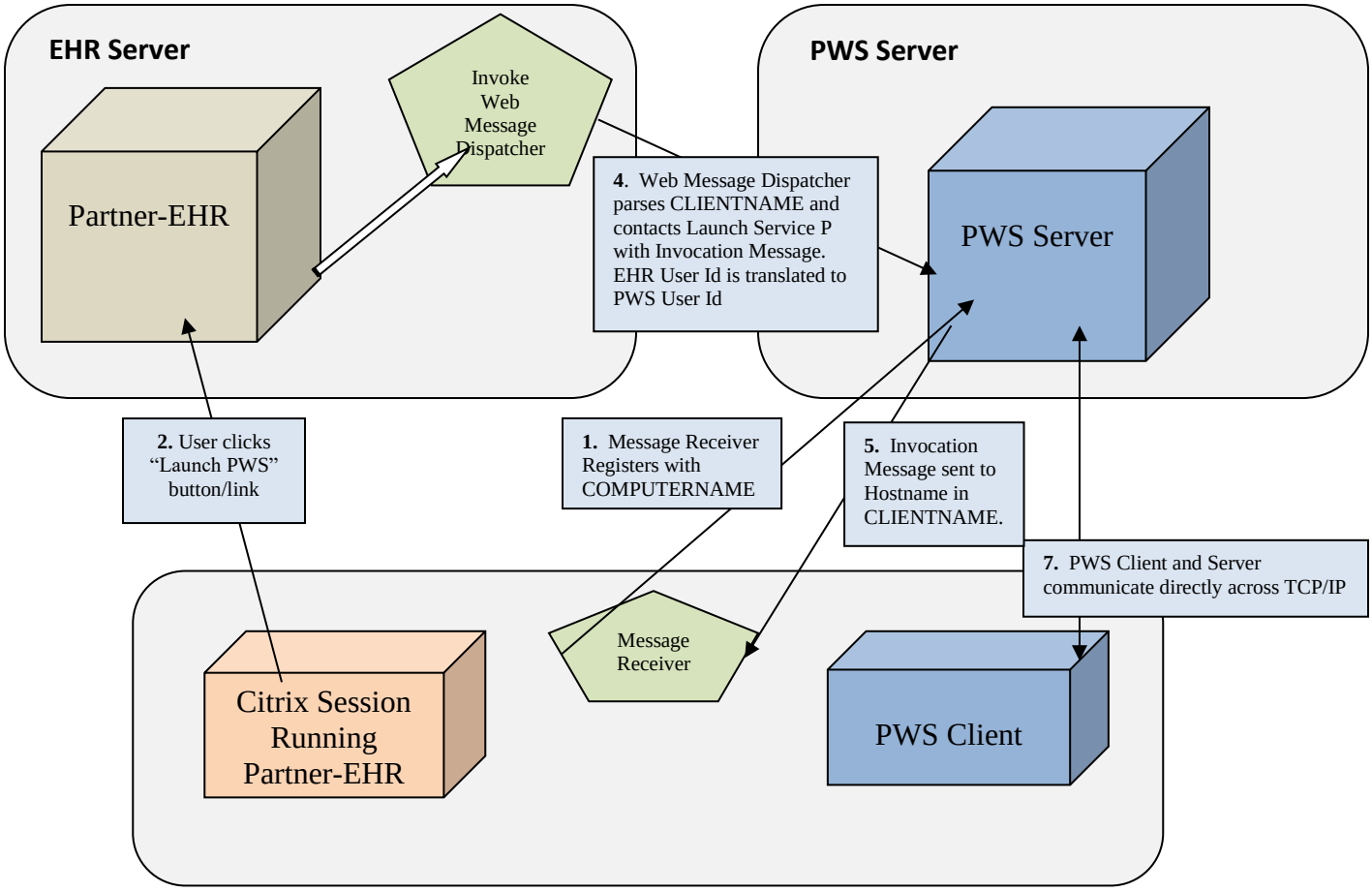
.

1.3.3 Command Line Invocation using PeriWatch Surveillance client



1.3.4 Web Invocation



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1.4 User Mapping configuration:

PWS usernames may differ from those of the partner-EHR. To accommodate such configuration, PWS provides a real-time translation service (step four above), whereby EHR userIDs are translated during the process of SSO launch. This service also applies to interfacing of, e.g., annotations inbound to PWS from the partner-EHR.

Users at configured sites will need to provide the PWS Integrations Team representative with a list of users authorized to use the SSO feature. Typically, this takes the form of an Excel spreadsheet with two columns:

EHRuserID1 PWSuserID1
EHRuserID2 PWSuserID1
EHRuserID3 PWSuserID2

Note that multiple EHR userIDs may be mapped to a single PWS userID. Upon receipt of a launch request (or interface message) containing EHRuserIDx, PWS agents will query a database table for the corresponding PWS userID and perform an inline substitution.

Maintenance of this lookup table requires submission of new userID pairs to the PWS Integrations Team.

1.5 Installation/Configuration Detail

1) Obtain a copy of the Message Dispatcher files from a configured PWS Server. A server that has already been upgraded and set up for operation of the PeriWatch system.
2) The database is updated with user cross-mapping.
3) For the command line invocation using Message Dispatcher implementation nothing further must be done to the Message Dispatcher, other than having it available for launch by the EHR software when a button is clicked on the EHR software by the user.
4) Changes for the file-watch invocation on the PWS server side:
    a. The location of the directory where the files will be deposited by the EHR software.
    b. Whether there will be one instance of Message Dispatcher software running for the entire EHR application server or not. This decision is necessary because with the file-based mode the Message Dispatcher must run continuously to react to file creation.
    c. The File based approach requires that the hostname be available so that messages can be embedded. In the case that there is only one instance of Message Dispatcher running for multiple EHR sessions, the hostname would have to be communicated via either the filename or embedded in the directory name that is being watched (perhaps one directory per hostname).

    After all these decisions are made PeriGen operations and Tech Support will set up the Launch Service configurations appropriately in a custom configuration file residing on the PeriWatch Server. The Message Dispatcher has been modified to request and use this configuration and set itself up appropriately.

5) The message receiver is pushed to client workstations.
6) Message Receiver configured to restart on startup of workstations.

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2 Maternal Interface Specification

The purpose of this section is to outline the PeriWatch Surveillance (**PWS**) expectations for HL7 message exchange with other systems. This document should form the foundation for interface design discussions and presumes prior knowledge of the Health Level Seven (**HL7**) messaging standard; details available at http://www.hl7.org.

With this version of PeriWatch Surveillance, the following logical datasets are expected to be in scope for each implementation of the system. Specific scope is to be determined during joint clinical workflow mapping exercise. Please refer to detail mapping templates in the appendix:

- ADT Inbound
- Results (ORU)
    - Documented results (Inbound/Outbound)
    - Labs (Inbound)
    - Outcomes (Inbound)
- Meds Inbound (RAS)

2.2 Communications

The PWS integrations engine resides on PWS training and production application servers, there sharing server IPs, network domain authorization/restrictions, and server hardware and database with the related application. As hardware and database are shared between application and integrations engine, PWS expresses strong preference towards the configuration of a **client-side filter**, where possible only passing messages related to existing or potential PWS patients to our engine for consideration and consumption. Often this filtering is accomplished by logic parsing PV1.3.1, passing messages only for patients with an assigned location of (e.g.) LND, LDRP, and/or OP OB Clinic; specific implementation of these filters is at client discretion. Additional requisite filtering may be required; details below where relevant.

Interface messaging occurs across TCP/IP connections. Our standard implementation port for incoming messages is **17676**. Outbound message destinations are expected to be the IP and port of a receiving interface engine; however, data may also be placed on a local file structure in various formats for collection at client preference.

Splitting of inbound message types occurs within the engine, and therefore messages should be collected at client systems and transmitted to PWS in a unitary stream where multiple system are involved (e.g., collect ADT, Lab, and Pharmacy messages for serial transmission to PWS on port 17676). Due to the loosely-coupled nature of the integrations engine with the application, messages need not be properly sequenced to be properly processed (e.g., reception of a lab message for a patient for which we have not yet received an ADT message is acceptable; reception of an ADT update message prior to reception of an ADT admit message is acceptable).

PWS is configured for serialized/bimodal message handling, transmitting or waiting for reception of acknowledgment messages immediately after message reception. Failed transmissions will retry or wait for retry until five attempts have passed, then closing the port and marking the route in error.

The low-level format of the HL7 data stream adheres to standard:

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<SB>dddd**<EB><CR>**

…where **<SB>** is the start block character (1 byte ASCII <VT>, 0x0B),
dddd represents the HL7 message data stream,
<EB> is the end block character (1 byte ASCII <FS>, 0x1C), and
<CR> is the carriage return character (0x0D).

2.3 Charting Interface Description

Where a typical downstream ancillary will react to the reception of an ADT^A01 message by admitting that patient to their census, PWS will not take any action – instead the message contents are placed in a pool of data from which a user can draw information during a manual admission into the application. Similarly, pharmacy inbound messages are primarily for pre-population of medication information when users document an administration in our application. As this is a deviation from common expectation, feel free to ask for more detail.

HL7 Message Definitions

The PWS interface standard adheres to the published HL7 standards for any v2.x message type. Valid transmitted message segments which are not mentioned in the following message definitions are ignored by our application.

Message delimiters adhere to standard: |,^,&,~, and \ being reserved characters to delimit fields, components, subcomponents, repeating fields, and escaped characters (respectively).

PWS currently has the capacity to process inbound HL7 messages pertaining to registration (ADT), lab results (ORU), and Pharmacy messages (RDS, RDE, and RAS). Inbound data from host EHR systems are accepted as specially formatted ORU unsolicited result (R01) messages and/or inside OBX message segments appended to ADT messages. PWS also has facilities for the export of vitals and annotations, as well as metadata surrounding discharge and audit reporting. Provision of additional inbound or outbound message exchange is outside the scope of this document.

The following sections define the required and optional components of PWS HL7 messages. Indentations denote logical message sub-blocks; fields in curly braces (**{}**) may repeat, those in square brackets (**[]**) are optional per specification.

Message Structures

ADT^A0{1,2,3,4,5,8}	Admit, Transfer, Discharge, Registration, PreAdmit, Update
MSH	Message Header
EVN	Event Type
PID	Patient Identifier
NK1	Next-of-Kin / Emergency Contact
PV1	Patient Visit Information
[{ **PV2** }]	Additional Patient Visit Information

[{ **OBX** }]	Observation Segments
[{ **AL1** }]	Patient Allergy Information
[{ **DG1** }]	Patient Diagnoses

Admit, Discharge, Transfer (ADT)
PWS only processes received Admit, Transfers, Discharges, Registration, Pre-Admit, and Update (A01, A02, A03, A04, A05, and A08, respectively) messages.

A06, A07, A11 and A12 messages can be received/processes after being converted to appropriate message type (See below). All other message types are ignored and should be filtered from transmission.

Auto ADT, HL7 Trigger	Note
A06 - O/P to I/P	Sites to convert to an A02
A07 - I/P to O/P	Sites to convert to an A02
A11 - Cancel/Undo Admit	Sites to Convert to an A03
A12 - Cancel/Undo Transfer	Sites to Convert to an A02

Please see Appendix II for list of ADT fields transmitted to PWS.

ADT Example

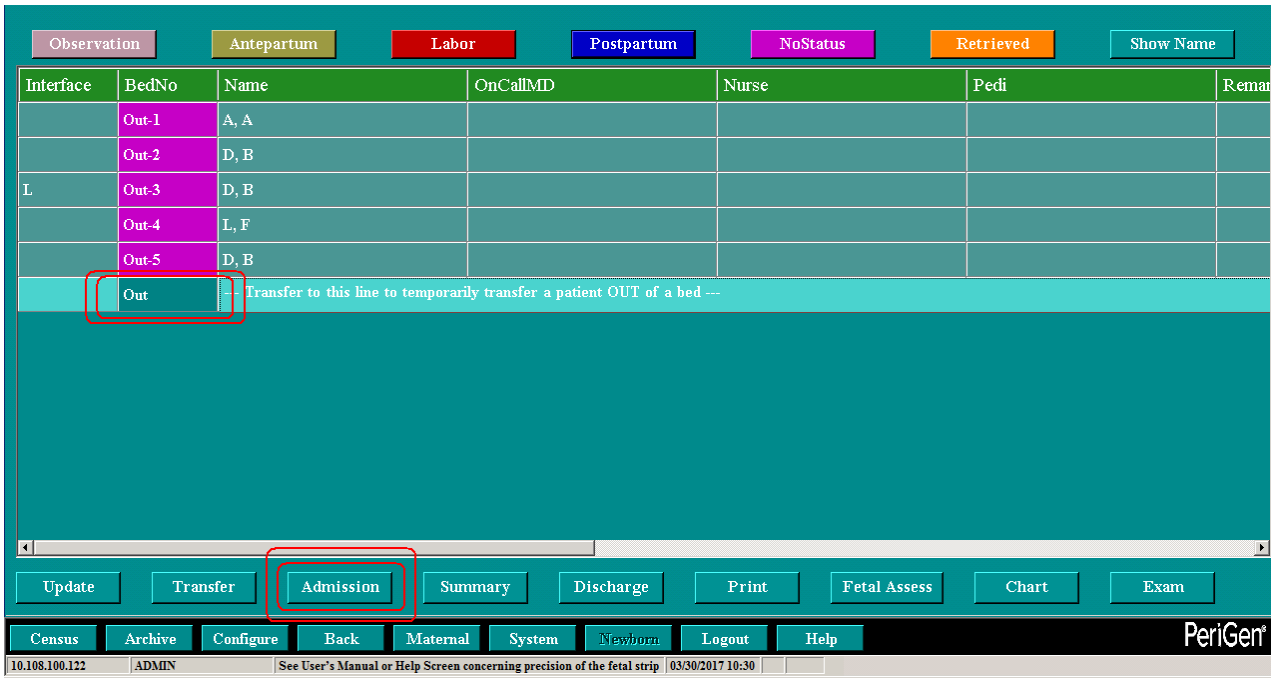
The following section describes the expected behavior of a “standard” sample of ADT and Lab HL7 messages.

The following message was run through the interface engine into PWS (message formatted for legibility):

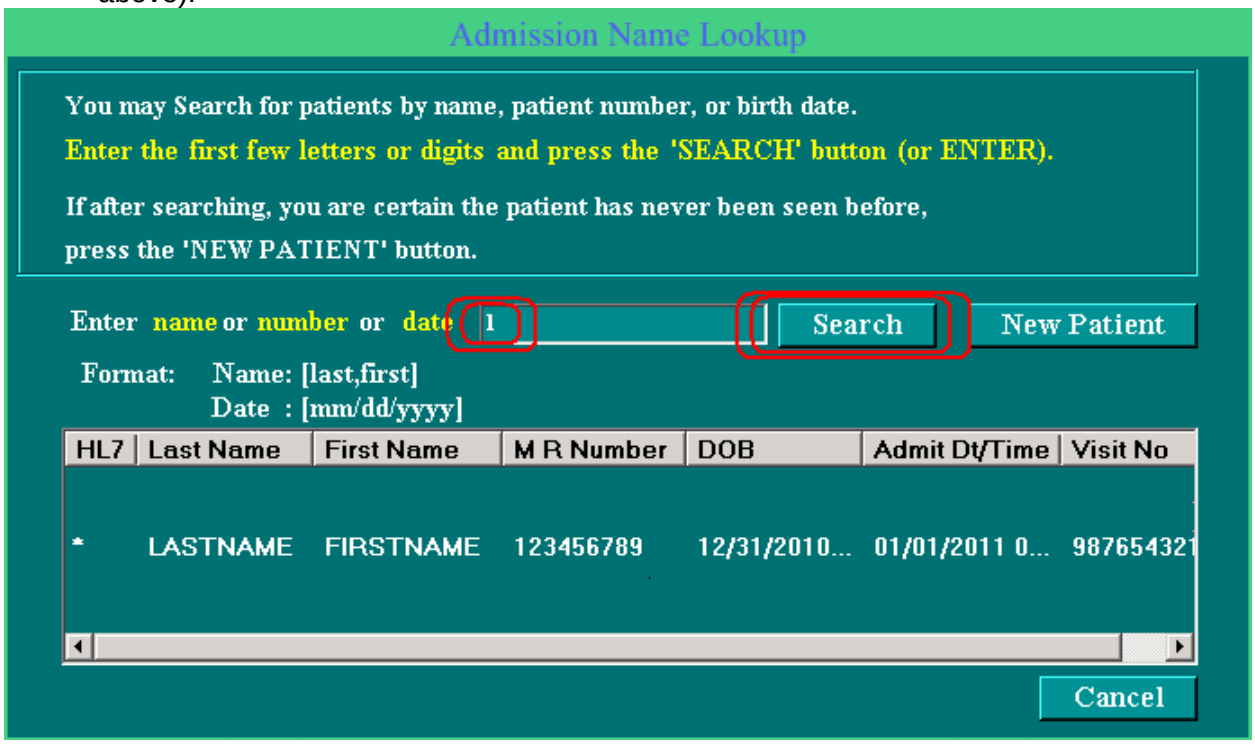
```
MSH|^~\\&|SendApp|SendFac|RecvApp|RecvFac|201101010001||ADT^A08|ADT201947277|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202||SIGN|SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)2020203^^^^^202^2020204
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|
```

To review the application’s consumption of this message:

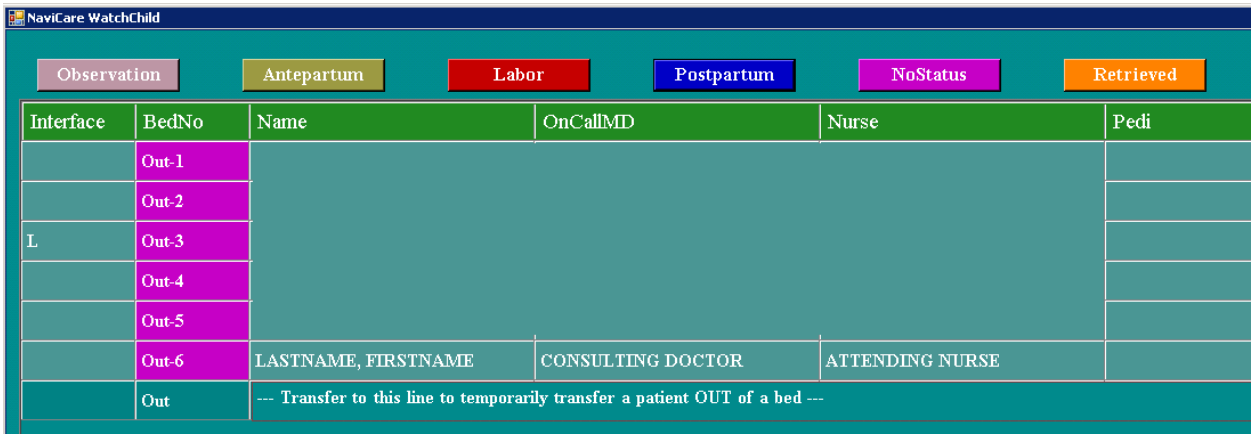
- Open the PWS application,
- Sign in, and
- Navigate to the census screen:



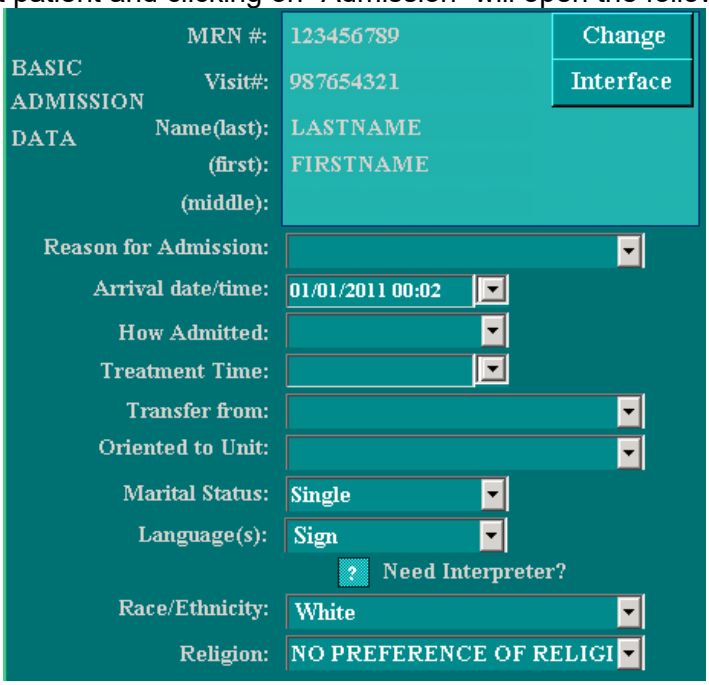
- Select a bed into which you can admit the patient and click on “Admission” (as shown above).



- Search for your patient based on some known value (above, searching for “1” as part of the Name, MRN, or DateTime)
- Click “Search”
- In the resulting list, patients whose information has been obtained from inbound HL7 ADT messages will have an ***** in the “HL7” column, as above. Select the desired patient.

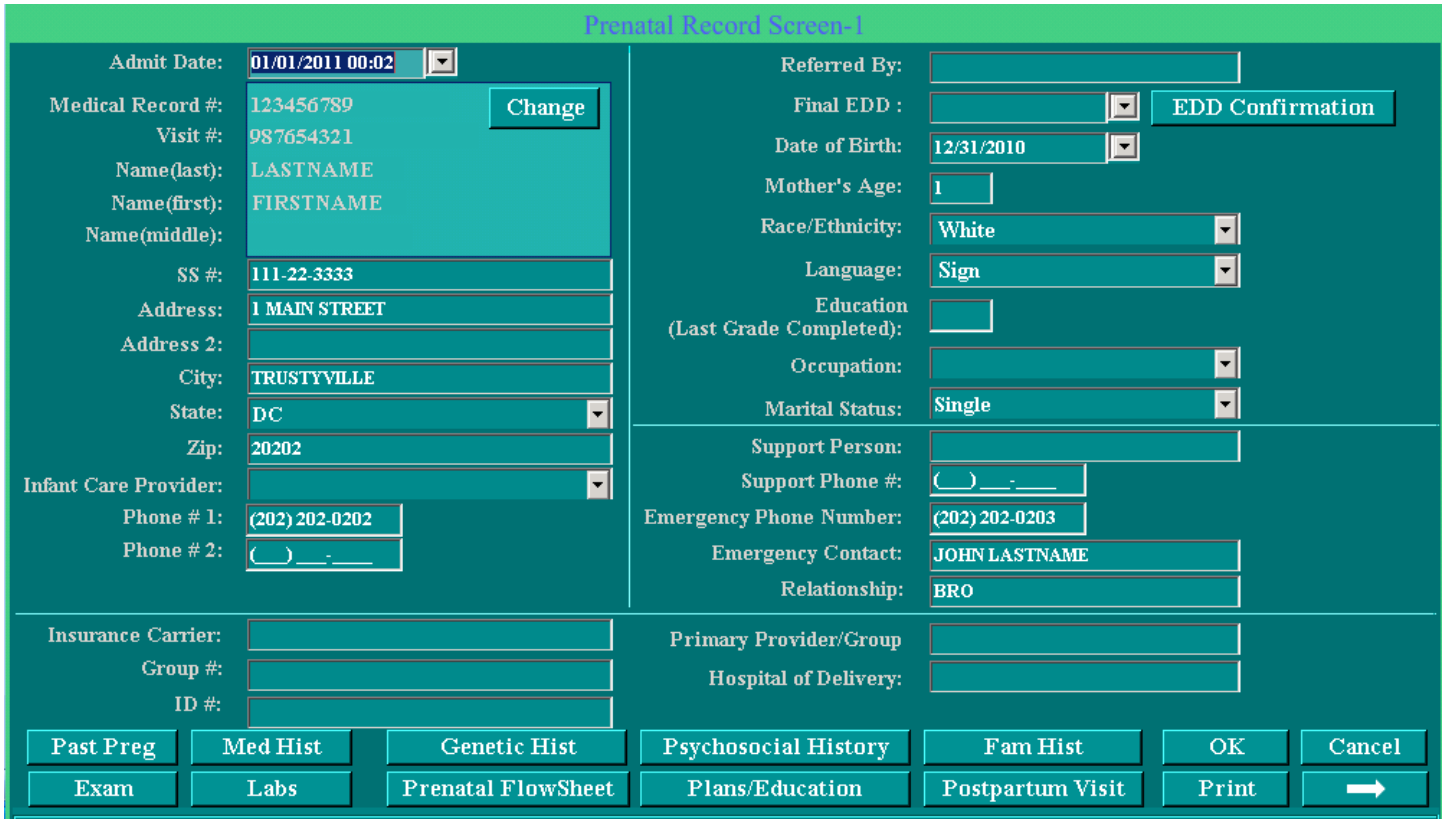


- The census screen will now have your patient listed, as shown in the screen clip above. Selecting that patient and clicking on “Admission” will open the following screen:



- Notice the fields populated in the screen above; comparing them against the HL7 message at the top of this document, the data source and destination are apparent.
- Further data is found elsewhere; select again the patient from the census and click “Chart”

- Selecting “Prenatal“ reveals the following screen with imported data:



The majority of the Obstetric Admitting and Prenatal records can be populated via the HL7 interface. The above describes the “standard” capabilities of the interface, if additional pre-population is desired please contact your interfaces representative or PWS project manager.

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Outcomes Data

Mapping outcomes data is required in order to provide meaningful reports mapping early warning trends detected by the Surveillance/Vigilance platform and actual baby/maternal outcomes data elements such as delivery mode, APGAR scores etc.
Please refer to the EMR HL7 mapping template for the list of all ideal outcomes fields that could be relevant for effective correlation and reporting. It is possible that some fields are not readily available in the source system in which case they cannot be mapped. The scope is to be decided in collaboration with PeriGen clinical and interface team.

Laboratory Results
Results are received in unsolicited observation (ORU^R01) messages from LIS systems. This message type is reused for subsequent interface functionality; absent any requisite (and later-described) distinguishing information, received ORU^R01 messages are presumed to be laboratory observations/results.

ORU^R01	Unsolicited Results
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
{	
OBR	Observation Request
{	
OBX	Observation / Result
[{ **NTE** }]	Notes and Comments
[**SPM**]	Specimen
}	
}	

The following table outlines the default set of labs of interest to drive the analytics in the Vigilance platform:

LABs	**Urine**	**Microbiology**
WBC	Protein/Creatinine Ratio	Group B strep
Hemoglobin	Protein	Blood Cultures
HCT		
Differential		
creatinine		
ast		

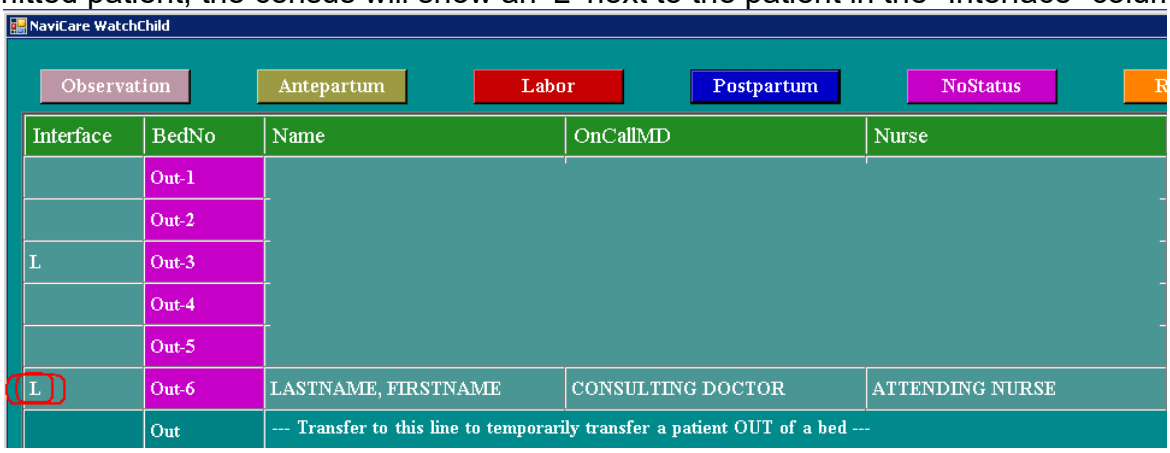
alt		
LDH		
Uric Acid		
PT/INR		
PTT		
Fibrinogen		
D-Dimer		

Lab Example

The following message was run through the interface engine into PWS (message formatted for legibility):

```
MSH|^~\\&|SendApp2|SendFac2|RecvApp|RecvFac|201101010021||ORU^R01|ADT201947278|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202|||SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)202-0203^^^^^202^2020204|
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|OUT/INPATIENT|AlternateVisitNo|||||||
   |||||||||||||||||201101010002|||||||005^NURSE^ATTENDING
OBR|1|ORDER_NO1|FILLERNO1^^|TESTCODE1^TESTDESCR^^^^|||201101010020||||||201101010021||||||||
   201101010022||||||
OBX|1|ST|RESULT_CODE^RESULTDESCR^||VALUE|MG/DL|LOW-HIGH|TERRIBLE|||||||||
NTE|1||****Comment Here****|
```

The PID segment of the passed message must match the MRN, first and last name, and date-of-birth of a patient who has either been manually admitted into PWS or for whom an ADT message exists and has been or will be sent to PWS. Upon arrival of a lab result message for an admitted patient, the census will show an 'L' next to the patient in the "Interface" column:



- Click on this "L" to show the Laboratory Results screen:

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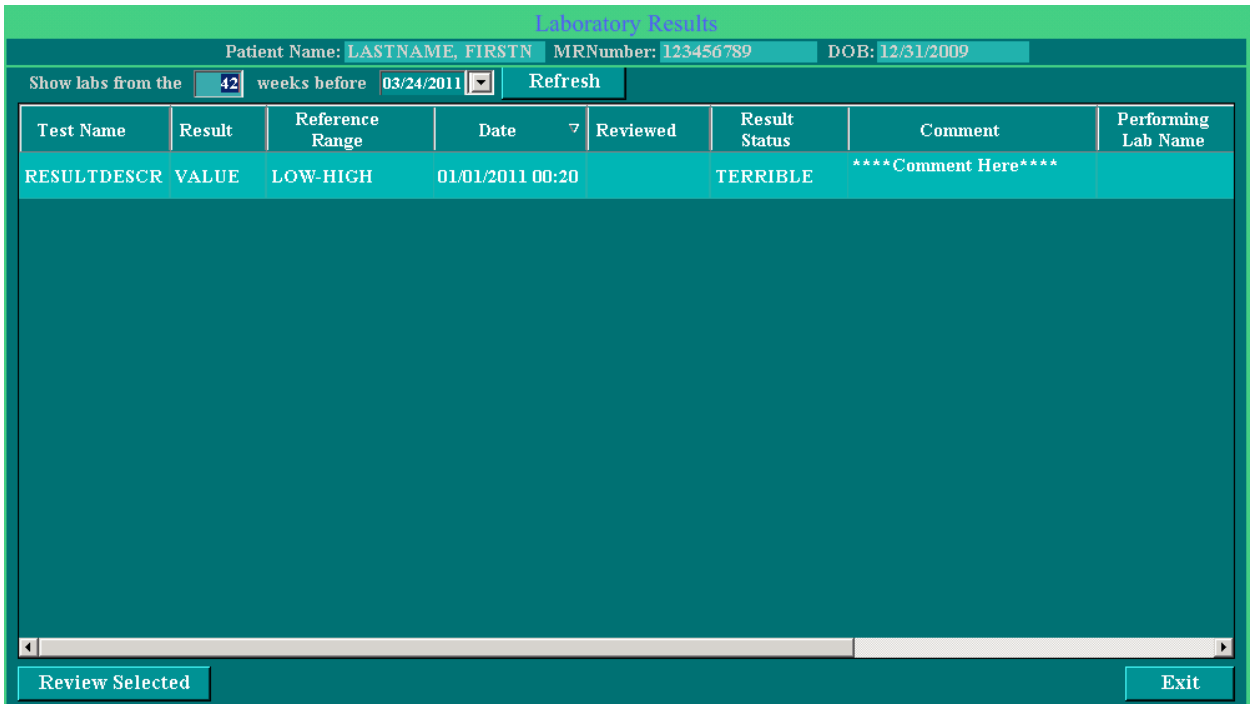

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Pharmacy:

RDE^O11	Encoded Order Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXE	Pharmacy Encoded Order
{ **RXR** }	Pharmacy/Treatment Route
RDS^O13	Dispense Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXD	Pharmacy/Treatment Dispense
{ **RXR** }	Pharmacy/Treatment Route
RAS^O17	Administration Messages
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
RXE	Pharmacy Encoded Order
{ **RXR** }	Pharmacy/Treatment Route

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Inbound Annotations
Inbound annotations are expected according to the previously described ORU^R01 message container.

Annotations themselves are to be passed in OBX segments (which may repeat). An annotation's unique identifiers are the patient's medical record number, the datetime of the annotation (with resolution fixed at "second", 120100.000 and 120100.999 would be interpreted as occurring at the same time), and the field prefix.

OBX|n||ANNOTATION EXAMPLE||SOME TEXT HERE.||||||F|||20121212121202||USER

…becomes a new annotation by user at 12:12:02 12Dec12 – "ANNOTATION EXAMPLE: SOME TEXT HERE."

DateTime of the annotation must be localtime of the PWS server, as we perform an internal conversion to UTC based on the PWS application's server time.

Inbound Annotation OBX Segment definition (all fields are required):

Outbound Vitals and Annotations
Outbound vitals and annotations are transmitted utilizing the previously described ORU^R01 container. OBR.4.2 will contain either "Vitals" or "Annotation", to differentiate the message contents.

For outbound vitals, OBX.3 will contain the vital sign being detailed; OBX.5 will hold the value and OBX.6 the unit – e.g.,

OBX|2|NM|NIBPSystolic||118|^mmHg|||||

Available vital signs include:

- Maternal Heart Rate
- SpO_2
- Maternal Mean, Systolic, and Diastolic Blood Pressure

Outbound annotations are similarly formatted, with OBX.3 equaling "Annotation" for all, and OBX.5 containing the annotation – e.g.,

OBX_n			Observation / Result
.3.1	Field Prefix	**.14**	Date/Time of the Annotation
$.5_0$	Annotation Text	**$.16_0.1$**	Responsible Observer ID
.11	Insert Mode { F = New, C = Update, D = Delete }		

OBX||TX|Annotation||Strip evaluated by RN.

Please see Appendix I for list of Annotations transmitted from PWS.

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Outbound Dynamic Data Distribution (3D) Report Metadata

Outbound dynamic data distribution metadata messages are transmitted utilizing the previously described ORU^R01 container.
Please see **Appendix I** for list of Uterine Fetal Assessment, Cervical Exam transmitted from PWS

Segment and Field Definitions

The following tables describe fields directly consumed by PWS. The intention of this section is to describe which fields are actually and directly mapped into (or out of) the PWS application. Fields required by the HL7 standard specification are still required for a message to be considered valid: omission of a field otherwise required by the HL7 standard specification from tables below indicates only that it is otherwise ignored by the PWS interface engine logic.

Unless otherwise noted, any additional subfield breakdowns will default to the first segment (e.g., if XXX.3.2 is noted in the table below but a given message defines subfields within XXX.3.2, XXX.3.2.1 will be utilized by default). If a field may repeat per specification, this field will be notated within the table by either a subscript **0** or **n;** indicating our use of only the first of the repeats and ignoring all subsequent fields or processing all passed repeats, respectively. Thus $AL1_n.5_0$ indicates that all passed allergy segments are processed, but that only the first .5 segment is considered for each AL1 – any subsequent .5 segments in a given AL1 message would be ignored.

All fields are considered of type **String** save **DateTime** fields, which are considered strings but parsed into MM/DD/YYYY HH:MM:SS.

Field Description	Field in HL7 message	Required	Optional	Ignored
	MSH			
Field Separator	MSH.1	X		
Encoding Characters	MSH.2	X		
Sendiing Application	MSH.3	X		
Sending Facility	MSH.4	X		
Receiving Application	MSH.5	X		
Receiving Facility	MSH.6	X		
Date/Time of Message	MSH.7	X		
Security	MSH.8			X
Message Type	MSH.9	X		
Message Control ID	MSH.10	X		
Processing ID	MSH.11	X		
HL7 Version ID	MSH.12	X		
Sequence Number	MSH.13			X
Continuation Pointer	MSH.14			X
Accept Ack Type	MSH.15	X		

Field Description	Field in HL7 message	Required	Optional	Ignored
Application App Type	MSH.16			X
Country Code	MSH.17			X
	EVN			
Set ID	EVN.1			X
Date/Time of Event	EVN.2	X		
Date/Time Planned Event	EVN.3			X
Event Reson Code	EVN.4			X
Operator ID	EVN.5			X
	PID			
Set ID	PID.1	X		
Patient ID (External)	PID.2			X
Patient ID (Internal)	PID.3	X		
Alternate Patient ID	PID.4			X
Patient Name	PID.5	X		
Mothers Maiden Name	PID.6			X
Patients DOB	PID.7	X		
Patient Sex	PID.8		X	
Patient Alias	PID.9			X
Race	PID.10		X	
Patient Address	PID.11		X	
Country Code	PID.12		X	
Home Phone	PID.13		X	
Business Phone	PID.14		X	
Patient Language	PID.15		X	
Religion	PID.17		X	
Patient Full Account Number	PID.18	X		
Patient SSN	PID.19		X	
Patient Drivers license	PID.20			X
Mothers Identifier	PID.21			X
Ethnic Group	PID.22		X	
Birth Place	PID.23			X
Multiple birth indicator	PID.24			X
Birth Order	PID.25			X
Citizenship	PID.26			X
Veteran Military Status	PID.27			X
	PV1			

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Field Description	Field in HL7 message	Required	Optional	Ignored
Set ID	PV1.1	X		
Assigned Patient Location	PV1.3	X		
Prior Patient Location	PV1.6	X (A02)		
Consulting Doctor	PV1.9		X	
Admitting Doctor	PV1.17		X	
Visit Number	PV1.19	X		
Admission Date/Time	PV1.44	X		
Discharge Date Time	PV1.45	X (A03)		
Attending Nurse	PV1.52		X	
	PV2			
Set ID	PV2.1			X
Reason for Admission	PV2.3			X
	NTE			
Set ID	NTE.1		X	
Source of Comment	NTE.2		X	
Notes/Comments	NTE.3		X	
Comment Type	NTE.4		X	
	AL1			
Set ID	AL1.1		X	
Allergen Type Code	AL1.2		X	
Allergen Code/Mnemonic/description	AL1.3		X	
Allergen Severity Code	AL1.4			X
Allergen Reaction Code	AL1.5			X
Identification Date	AL1.6			X
	NK1			
Set ID	NK1.1		X	
Contact Name	NK1.2		X	
Relationship	NK1.3		X	
Phone Number	NK1.5		X	

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3 PeriWatch Newborn Interface Specification

The purpose of this document is to outline the PeriWatch Surveillance (**PWS**) expectations for HL7 message exchange with other systems. This document should form the foundation for interface design discussions and presumes prior knowledge of the Health Level Seven (**HL7**) messaging standard; details available at http://www.hl7.org.

3.2 Communications

The PWS integrations engine resides on PWS training and production application servers, there sharing server IPs, network domain authorization/restrictions, and server hardware and database with the related application. As hardware and database are shared between application and integrations engine, PWS expresses strong preference towards the configuration of a **client-side filter**, where possible only passing messages related to existing or potential PWS patients to our engine for consideration and consumption. Often this filtering is accomplished by logic parsing PV1.3.1, passing messages only for patients with an assigned location of (e.g.) LDRP, NICU; specific implementation of these filters is at client discretion. Additional requisite filtering may be required; details below where relevant.

Interface messaging occurs across TCP/IP connections. Our standard implementation port for incoming messages is **17676**.

Splitting of inbound message types occurs within the engine, and therefore messages should be collected at client systems and transmitted to PWS in a unitary stream where multiple systems are involved (e.g., collect ADT and Lab messages for serial transmission to PWS on port 17676). Due to the loosely-coupled nature of the integrations engine with the application, messages need not be properly sequenced to be properly processed (e.g., reception of a lab message for a patient for which we have not yet received an ADT message is acceptable; reception of an ADT update message prior to reception of an ADT admit message is acceptable).

PWS is configured for serialized/bimodal message handling, transmitting or waiting for reception of acknowledgment messages immediately after message reception/transmission. Failed transmissions will retry or wait for retry until five attempts have passed, then closing the port and marking the route in error.

The low level format of the HL7 data stream adheres to standard:

<SB>dddd**<EB><CR>**

…where **<SB>** is the start block character (1 byte ASCII <VT>, 0x0B),
dddd represents the HL7 message data stream,
<EB> is the end block character (1 byte ASCII <FS>, 0x1C), and
<CR> is the carriage return character (0x0D).

3.3 Charting Interface Description

Where a typical downstream ancillary will react to the reception of an ADT^A01 message by admitting that patient to their census, PWS will not take any action – instead the message contents are placed in a pool of data from which a user can draw information during a manual admission into the application. As this is a deviation from common expectation, feel free to ask for more detail.

3.4 HL7 Message Definitions

The PWS interface standard adheres to the published HL7 standards for any v2.x message type through v2.6. Valid transmitted message segments which are not mentioned in the following message definitions are ignored by our application.

Message delimiters adhere to standard: |,^,&,~, and \ being reserved characters to delimit fields, components, subcomponents, repeating fields, and escaped characters (respectively).

PeriWatch Newborn currently has the capacity to process inbound HL7 messages pertaining to registration (ADT) and lab results (ORU). Provision of additional inbound or outbound message exchange is outside the scope of this document.

The following sections define the required and optional components of PWS HL7 messages. Indentations denote logical message sub-blocks; fields in curly braces (**{}**) may repeat, those in square brackets (**[]**) are optional per specification.

3.5 Message Structures

ADT^A0{1,4,5,8}	Admit, Registration, PreAdmit, Update
MSH	Message Header
EVN	Event Type
PID	Patient Identifier
NK1	Next-of-Kin / Emergency Contact
PV1	Patient Visit Information
[{ **PV2** }]	Additional Patient Visit Information
[{ **OBX** }]	Observation Segments
[{ **AL1** }]	Patient Allergy Information
[{ **DG1** }]	Patient Diagnoses

Admit, Discharge, Transfer (ADT)
PWS only processes received Admit, Registration, Pre-Admit, and Update (A01, A04, A05, and A08, respectively) messages; all others are ignored and should be filtered from transmission. PWS does not transmit any ADT messages.

3.6 ADT Example

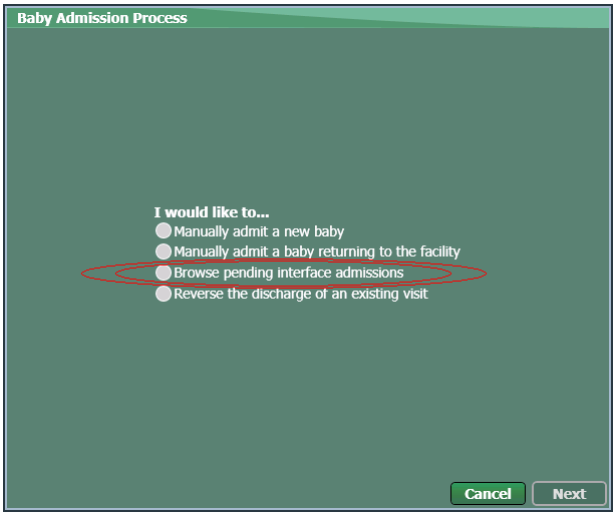
The following message was run through the interface engine into PWS (message formatted for legibility):

```
MSH|^~\\&|SendApp|SendFac|RecvApp|RecvFac|201101010001||ADT^A08|ADT201947277|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202||SIGN|SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
    NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)2020203^^^^^202^2020204
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|
   OUT/INPATIENT|AlternateVisitNo||||||||||||||||||||||||||201101010002||||||||005^NURSE^ATTENDING
```

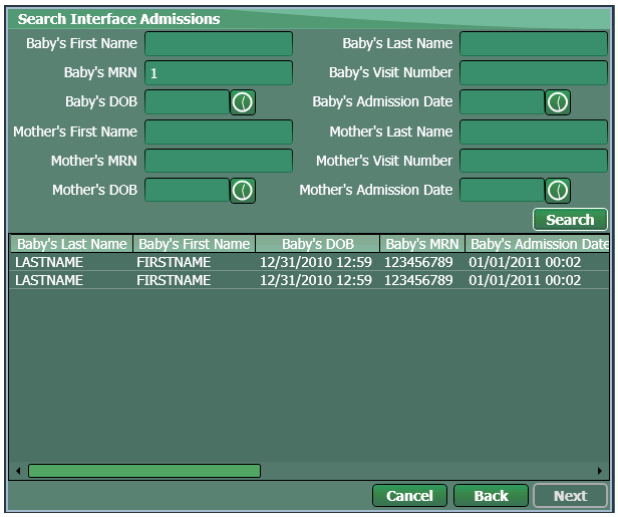
To review the application's consumption of this message:

- Open the PeriWatch Newborn application,
- Sign in, and
- Navigate to the census screen.

- Select “Admit” from the lower-left, then “Browse pending interface admissions”.



- Search for your patient based on some known value (above, searching for “1” as part of the Name, MRN, or DateTime); if you do not want to constrain the search by Baby’s Admission Date, you must select the text in the field and hit “Delete” or “Backspace” to clear the contents.
- Click “Search”



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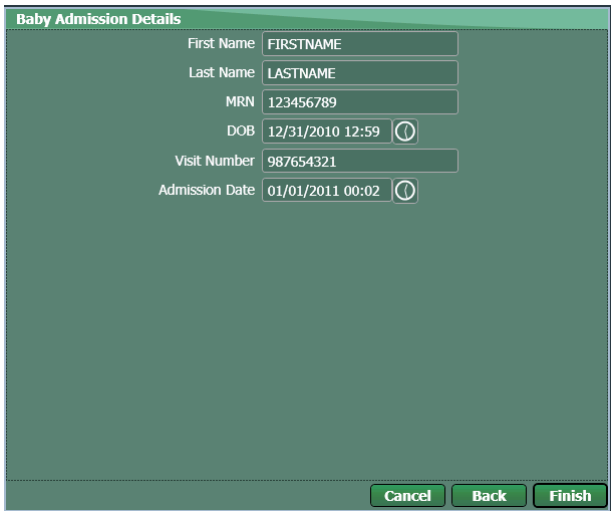
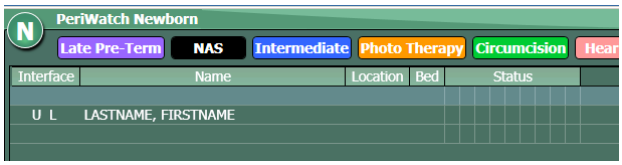
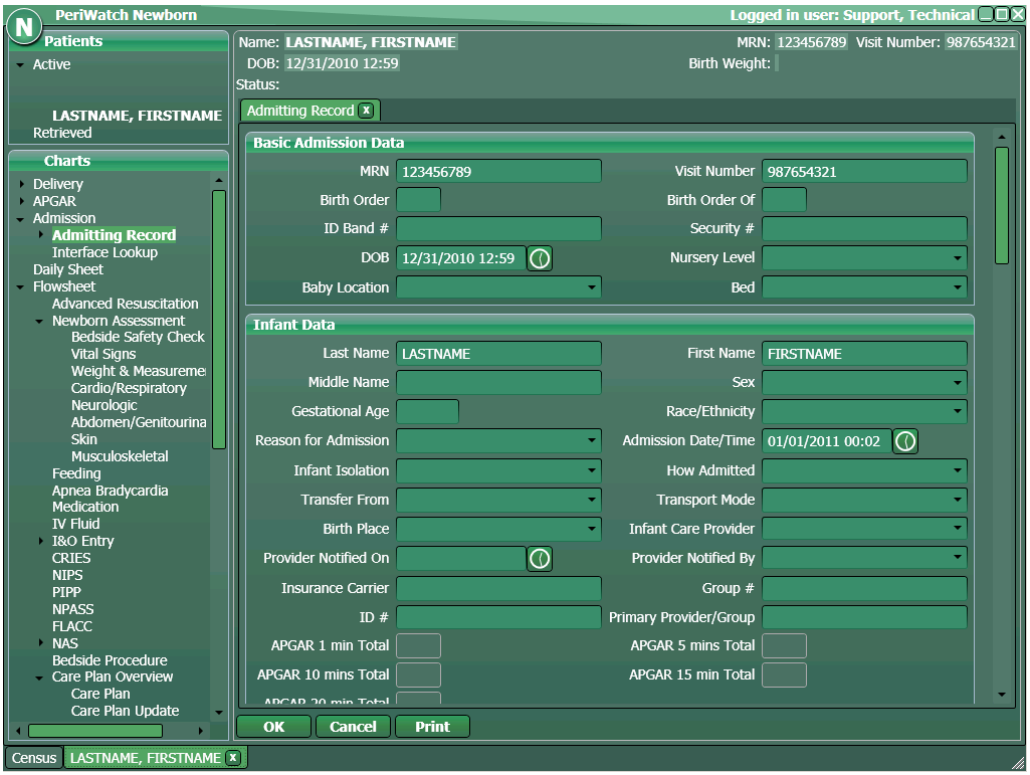
- In the resulting list, patients whose information has been obtained from inbound HL7 ADT messages will be returned. You will see one line in the table for each message received: unlike the maternal application, a single patient may have many results returned here. Selection of the top-most entry will generally return the most up-to-date and complete message. Select the desired patient.



- Verify the retrieved information and click Finish when complete.



- The census screen will now have your patient listed, as shown in the screen clip above. Selecting that patient and clicking on “Chart”, then “Admission” will open the following screen:



- Notice the fields populated in the screen above; comparing them against the HL7 message at the top of this document, the data source and destination are apparent.
- Where new ADT messages have arrived for a patient, click on the “U” in the census next to their name. On the resulting search screen, uncheck Admit Date Time and Date of Birth search options and click “Search”, to review new messages for incorporation into the patient record.

ORU^R01	Unsolicited Results
MSH	Message Header
PID	Patient Identifier
PV1	Patient Visit Information
ORC	Common Order
{	
OBR	Observation Request
{	
OBX	Observation / Result
[{ **NTE** }]	Notes and Comments
[**SPM**]	Specimen
}	
}	

Laboratory Results

Results are received in unsolicited observation (ORU^R01) messages from LIS systems. This message type is reused for subsequent interface functionality; absent any requisite (and later-described) distinguishing information, received ORU^R01 messages are presumed to be laboratory observations/results.

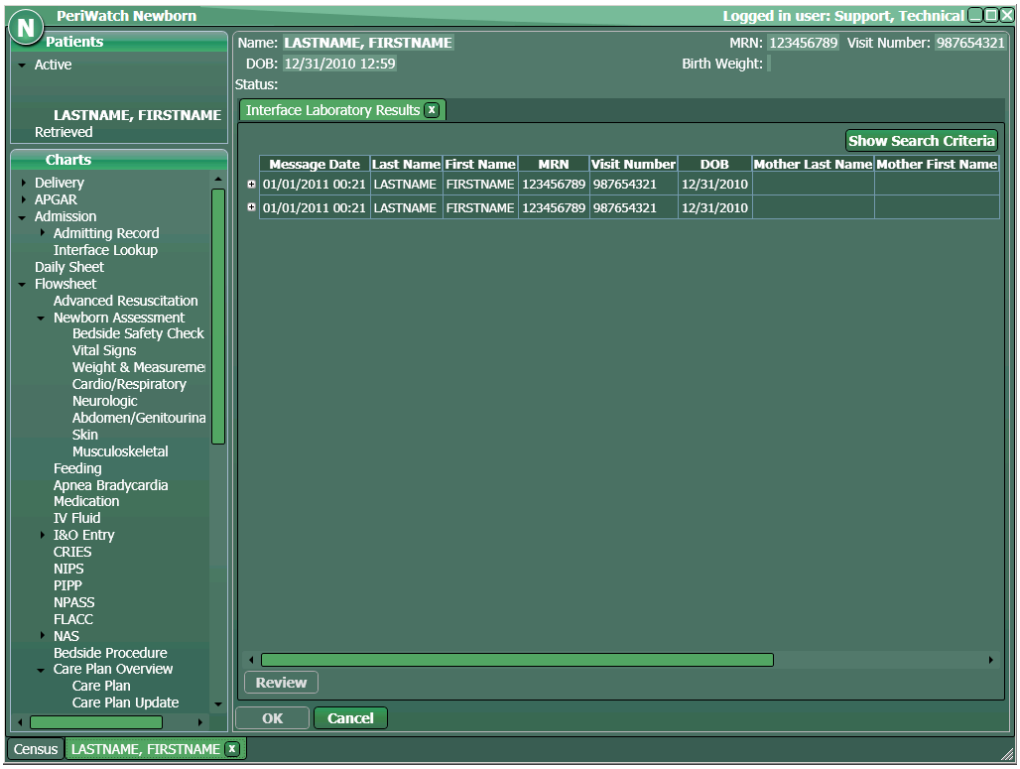
3.7 Lab Example

The following message was run through the interface engine into PWS (message formatted for legibility):

```
MSH|^~\\&|SendApp2|SendFac2|RecvApp|RecvFac|201101010021||ORU^R01|ADT201947278|P|2.3|||AL||||
PID|1||123456789||LASTNAME^FIRSTNAME||200112311259|M||WHITE|1 MAIN
   STREET^^TRUSTYVILLE^DC^20202||(202)202-0202^^^^^202^2020202|||SINGLE|NO PREFERENCE OF
   RELIGION|987654321|111-22-3333||||||||||
NK1|1|LASTNAME^JOHN|BRO|2 MAIN STREET^^TRUSTYVILLE^DC^20202|(202)202-0203^^^^^202^2020204|
PV1|1|I|UNIT^ROOM^BED^.||||001^DOCTOR^ATTENDING|002^DOCTOR^REFERRING|003^DOCTOR^CONSULTING|
   SERVICE||||ADMITSOURCE (e.g., ED)|||004^DOCTOR^ADMITTING|OUT/INPATIENT|AlternateVisitNo||||||
   |||||||||||||||||201101010002|||||||005^NURSE^ATTENDING
OBR|1|ORDER_NO1|FILLERNO1^^|TESTCODE1^TESTDESCR^^^^|||201101010020||||||201101010021|||||||
   201101010022|||||||
OBX|1|ST|RESULT_CODE^RESULTDESCR^||VALUE|MG/DL|LOW-HIGH|TERRIBLE|||||||||
NTE|1||****Comment Here****|
```

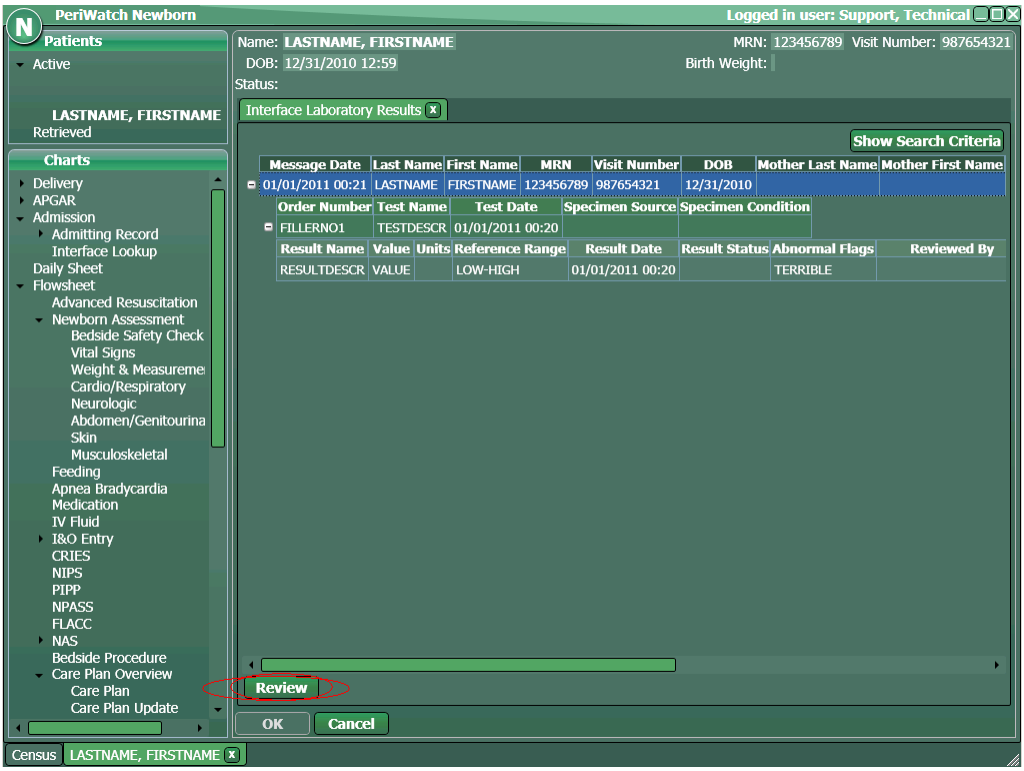
The PID segment of the passed message must match the MRN, first and last name, and date-of-birth of a patient who has either been manually admitted into PWS or for whom an ADT message exists and has been or will be sent to PWS. Upon arrival of a lab result message for an admitted patient, the census will show an 'L' next to the patient in the "Interface" column.

- Click on this "L" to show the Laboratory Results screen:

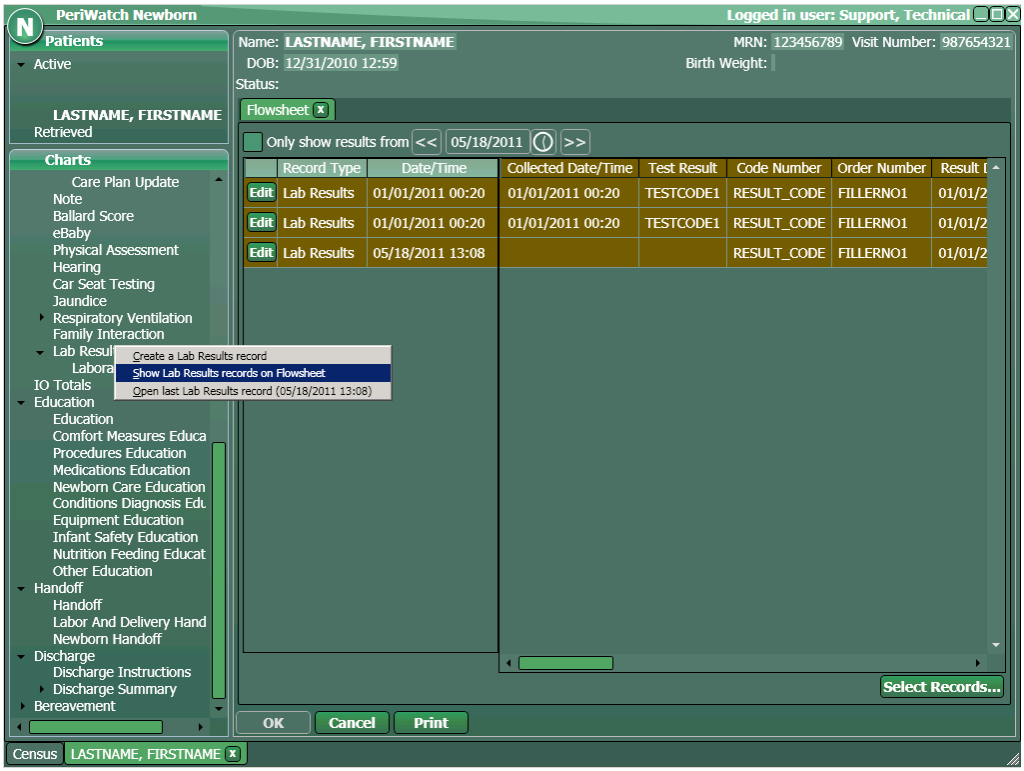


- There is a line for each inbound laboratory message – each message may contain one or many actual results. Expand the entries by clicking the ⊞ to see the results themselves; click on the

message row and select “Review” to add the lab result to the patient’s chart. (Lab messages for the patient will not be added to the patient’s medical record within PeriWatch Newborn unless reviewed.)



- To view results which have been reviewed and posted on the chart, right-click on “Lab Results” and select “Show Lab Results records on Flowsheet”


PeriWatch Newborn
Logged in user: Support, Technical
Patients
Active
LASTNAME, FIRSTNAME
Retrieved
Charts
Care Plan Update
Note
Ballard Score
eBaby
Physical Assessment
Hearing
Car Seat Testing
Jaundice
Respiratory Ventilation
Family Interaction
Lab Resul
Labora
IO Totals
Education
Education
Comfort Measures Educa
Procedures Education
Medications Education
Newborn Care Education
Conditions Diagnosis Edu
Equipment Education
Infant Safety Education
Nutrition Feeding Educat
Other Education
Handoff
Handoff
Labor And Delivery Hand
Newborn Handoff
Discharge
Discharge Instructions
Discharge Summary
Bereavement
Create a Lab Results record
Show Lab Results records on Flowsheet
Open last Lab Results record (05/18/2011 13:08)
Name: LASTNAME, FIRSTNAME
MRN: 123456789 Visit Number: 987654321
DOB: 12/31/2010 12:59
Birth Weight:
Status:
Flowsheet
Only show results from << 05/18/2011 >>
Record Type
Date/Time
Collected Date/Time
Test Result
Code Number
Order Number
Edit Lab Results 01/01/2011 00:20 01/01/2011 00:20 TESTCODE1 RESULT_CODE FILLERNO1 01/01/2
Edit Lab Results 01/01/2011 00:20 01/01/2011 00:20 TESTCODE1 RESULT_CODE FILLERNO1 01/01/2
Edit Lab Results 05/18/2011 13:08 RESULT_CODE FILLERNO1 01/01/2
Select Records...
OK
Cancel
Print
Census
LASTNAME, FIRSTNAME

3.8 Segment and Field Definitions

The following tables describe fields directly consumed by PWS. The intention of this section is to describe which fields are actually and directly mapped into (or out of) the PWS application. Fields required by the HL7 standard specification are still required for a message to be considered valid: omission of a field otherwise required by the HL7 standard specification from tables below indicates only that it is otherwise ignored by the PWS interface engine logic.

Unless otherwise noted, any additional subfield breakdowns will default to the first segment (e.g., if XXX.3.2 is noted in the table below but a given message defines subfields within XXX.3.2, XXX.3.2.1 will be utilized by default). If a field may repeat per specification, this field will be notated within the table by either a subscript **0** or **n;** indicating our use of only the first of the repeats and ignoring all subsequent fields or processing all passed repeats, respectively. Thus $AL1_n.5_0$ indicates that all passed allergy segments are processed, but that only the first .5 segment is considered for each AL1 – any subsequent .5 segments in a given AL1 message would be ignored.

All fields are considered of type **String** save **DateTime** fields, which are considered strings but parsed into the XML-defined UTC DateTime format (with GMT-offset presumed equal to that of the host EHR system).

Concerning the capture of an infant's mother's information: $PID.21_0.1$ (Mother's MRN) is populated for infants where available. Where NK1.3 (Relationship) is "MTH" or "MOTHER", the information from this segment is used to prepopulate information regarding the infant's mother. Currently First and Last Name are the only captured values; additional data capture may be possible upon request.

Field Description	Field in HL7 message	Required	Optional	Ignored
	MSH			
Field Separator	MSH.1	X		
Encoding Characters	MSH.2	X		
Sending Application	MSH.3	X		
Sending Facility	MSH.4	X		
Receiving Application	MSH.5	X		
Receiving Facility	MSH.6	X		
Date/Time of Message	MSH.7	X		
Security	MSH.8			X
Message Type	MSH.9	X		
Message Control ID	MSH.10	X		
Processing ID	MSH.11	X		
HL7 Version ID	MSH.12	X		
Sequence Number	MSH.13			X
Continuation Pointer	MSH.14			X
Accept Ack Type	MSH.15	X		
Application App Type	MSH.16			X
Country Code	MSH.17			X
	EVN			
Set ID	EVN.1			X

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Field Description	Field in HL7 message	Required	Optional	Ignored
Date/Time of Event	EVN.2	X		
Date/Time Planned Event	EVN.3			X
Event Reason Code	EVN.4			X
Operator ID	EVN.5			X
	PID			
Set ID	PID.1	X		
Patient ID (External)	PID.2			X
Patient ID (Internal)	PID.3	X		
Alternate Patient ID	PID.4			X
Patient Name	PID.5	X		
Mother's Maiden Name	PID.6			X
Patients DOB	PID.7	X		
Patient Sex	PID.8		X	
Patient Alias	PID.9			X
Race	PID.10		X	
Patient Address	PID.11		X	
Country Code	PID.12		X	
Home Phone	PID.13		X	
Business Phone	PID.14		X	
Patient Language	PID.15		X	
Religion	PID.17		X	
Patient Full Account Number	PID.18	X		
Patient SSN	PID.19		X	
Patient Driver's license	PID.20			X
Mothers Identifier	PID.21			X
Ethnic Group	PID.22		X	
Birthplace	PID.23			X
Multiple birth indicator	PID.24			X
Birth Order	PID.25			X
Citizenship	PID.26			X
Veteran Military Status	PID.27			X
	PV1			
Set ID	PV1.1	X		
Assigned Patient Location	PV1.3	X		
Consulting Doctor	PV1.9		X	
Admitting Doctor	PV1.17		X	
Visit Number	PV1.19	X		
Admission Date/Time	PV1.44	X		
Attending Nurse	PV1.52			X

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4 Appendix I

4.2 Strip Annotations:

Field	Value
Intrauterine resuscitation	Position changed IV bolus initiated- See MAR Oxygen at 10L/min via non-rebreather mask Called for assistance Pitocin off- See MAR Tocolytics given- See MAR Ultrasound to verify fetal heart rate
Fetal interventions	Scalp stimulation between contractions Ultrasound to verify fetal heart rate RN continuously at bedside evaluating FHR Modified pushing effort Vibroacoustic stimulation Amnioinfusion started Amnioinfusion stopped Amnioinfusion fluid returned
Membrane	Intact SROM- See EMR AROM- See EMR Bulging
Monitor	EFM applied Toco applied EFM adjusted Toco adjusted FSE applied FSE discontinued FHR via doppler FHR via auscultation Telemetry/wireless applied IUPC inserted IUPC zeroed IUPC flushed

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Field	Value
	IUPC discontinued
I&O Interventions	Ice chips given-Document volume in I/O Popsicle given-Document volume in I/O Clear liquids given-Document volume in I/O Up to bathroom Bedpan given- Document volume in I/O Straight catheter- Document volume in I/O PeriCare given Foley catheter inserted- Document volume in I/O Foley catheter discontinued- Document volume in I/O
IV	IV started Labs drawn with IV start IV bolus started- See MAR IV bolus stopped- See MAR New IV bag- See MAR IV saline locked IV fluids discontinued- See MAR
Induction/Augmentation	Pitocin started- See MAR Pitocin increased- See MAR Pitocin decreased- See MAR Pitocin discontinued- See MAR Cervical ripening agent- See MAR Laminaria Cervical ripening balloon placed Cervical ripening balloon discontinued
Positions	Birthing ball Chair Hands/knees High Fowlers Jet Hydrotherapy Knee chest Left lateral Left tilt Low Fowlers Peanut ball Right lateral Right tilt Rocking chair

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Field	Value
	Semi-fowlers Shower Sitting Squatting Standing Supine Tailor sitting Trendelenburg Walking Other
Comfort Measures	Underpad changed Coached with breathing/relaxation techniques Up to shower Assist to tub Laboring in tub Back rub given Hot pack applied Cold pack applied Family support at bedside Aromatherapy in use Doula support at bedside Hypnotherapy
Medication	Terbutaline given- See MAR Ephedrine given- See MAR Medication given- See MAR Magnesium sulfate bolus started- See MAR Magnesium sulfate bolus complete- See MAR Magnesium sulfate maintenance dose started- See MAR Magnesium sulfate maintenance dose discontinued- See MAR
Pushing Preparation	Patient reports feeling pressure Foley removed- Document volume in I/O Family at bedside Doula at bedside RN continually at bedside evaluating FHR Provider at bedside to push with patient Leg support by family/significant other Patient instructed in pushing technique
Second stage position	Semi-fowlers High fowlers

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Field	Value
	Right lateral Left lateral Squatting Hands/knees Knee chest Squatting bar In stirrups Pushing on side with leg up in stirrup Birthing bar Support of significant other Pushing with sheet around birthing bar
Pushing Effectiveness	Patient pushing well Patient pushing ineffectively Patient unable to feel pressure to push Baby crowning Making slow progress Epidural pump off
Delivery Preparation	Provider called to room for delivery Patient set up for delivery Perineal prep done Provider at bedside for delivery Second RN/OB Tech called for delivery NICU called for delivery NICU at bedside Anesthesia called for perineal dose Anesthesia at bedside
Labor provider name/title	<Free Text>
Method of notification	Electronic page Text page Overhead page Provider phoned Left message Provider at bedside
Request	Evaluate remote Evaluate in person Attend delivery
Notification Reason	Decels

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Field	Value
	Fetal baseline change Variability change Membrane status Labor status Uterine activity Pain Maternal vital signs change Bleeding Lab/diagnostic study Patient request Status update
Notification Reason	Decels Fetal baseline change Variability change Membrane status Labor status Uterine activity Pain Maternal vital signs change Bleeding Lab/diagnostic study Patient request Status update
Response	Provider returned call Provider given report Provider to bedside Provider leaves bedside Provider in department Provider reviewed fetal monitor strip
Strip Annotation (free text)	<Free Text>

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4.3 Uterine Fetal Assessment:

Field	Value
Monitor	External Toco Internal IUPC Palpation None Other
Frequency	<Numeric><Text> Minutes
Contractions per 10 minutes	<Numeric><Text> Minutes
Duration	<Numeric><Text> Minutes
Intensity	Mild Mild/Moderate Moderate Moderate/Strong Strong Other
Intensity IUPC mmhg	<Numeric><Text>
Resting Tone	Mild Moderate Strong Soft Rigid Other
MVU	<Numeric><Text>
Characteristics	Regular Irregular Irritability Tachysystole Other
Patient Position	Birthing ball Chair Hands/knees High Fowlers Jet Hydrotherapy Knee chest Left lateral Left tilt Low Fowlers

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Field	Value
	Peanut ball Right lateral Right tilt Rocking chair Semi-fowlers Shower Sitting Squatting Standing Supine Tailor sitting Trendelenburg Walking Other
Comments	<Text>
Monitor	External Ultrasound Internal Fetal Spiral Electrode Auscultation Doppler Fetascope None Other
Presentation	Cephalic Cephalic Face Cephalic Brow Breech Breech Complete Breech Frank Breech Footling Shoulder Hand Compound Transverse Lie Oblique Lie Cord Unknown Other
Baseline Variability	Absent

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Field	Value
	Minimal (1-5 BPM) Moderate (6-25 BPM) Marked (>25 BPM) Unable to determine
Categories	Category I Category II Category III
FHR	<Text>
FHR Rhythym	Regular Irregular
Characteristics	Bradycardia Indeterminate Normal Sinusoidal Tachycardia
Accels	Present Absent
Decels	None Early Late Variable Prolonged Present
Reoccurrence	Recurrent Intermittent
Checklist	Met Not Met
Comments	<Text>
Membrane	Intact SROM AROM PROM PPROM Prolonged Rupture of Membranes Bulging Unknown
Fluid	Clear Cloudy

Field	Value
	Bloody Meconium Stained Meconium Thin Meconium Moderate Meconium Thick Yellow Brown Green Tinted Pinkish Unknown Other
Amount	<Text>
Rupture Date/Time	<Numeric><Text>

4.4 Cervical Exam:

Field	Value
Dilitation	0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5 5.5 6 6.5 7 7.5 8 8.5 9

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Field	Value
	9.5 10 0.5 Fingertip 9.5 Anterior Lip
Effacement	0 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Station	0 -1 -2 -3 -4 -5 1 2 3 4 5 Crown
Fetal Position	OA

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Field	Value
	LOA ROA OP LOP ROP OT LOT ROT
Cervical Position	Anterior Posterior
Consistency	Soft Medium Firm
Presentation	Cephalic Cephalic Face Cephalic Brow Breech Breech Complete Breech Frank Breech Footling Shoulder Hand Compound Transverse Lie Oblique Lie Cord Unknown
Comments	<Text>
Examined by	Physician/Nurse drop down

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5 Appendix II

5.2 OB Record Screen:

Field	Field in HL7 Message
MRN#	PID.3.1
Visit Number#	PID.18 OR PV1.19
LastName	PID.5.1.1
FirstName	PID.5.2
MiddleName	PID.5.3
Date of Birth	PID.7
Race/Ethnicity	PID.10
Languages	PID.15
Marital Status	PID.16
Religion	PID.17
SSN Number	PID.19
Physician/CNM	PV1.17
Reason for Admission	PV2.3
Emergency Name	NK1.2
Emergency Contact Number	NK1.5
Relationship	NK1.3
Bed Number	PV1.3.3

5.3 Prenatal Record Screen:

Field	Field in HL7 Message
MRN#	PID.3.1
Visit Number#	PID.18 OR PV1.19
LastName	PID.5.1.1
FirstName	PID.5.2
MiddleName	PID.5.3
Date of Birth	PID.7
Race/Ethnicity	PID.10
Languages	PID.15
Marital Status	PID.16
Religion	PID.17

Field	Field in HL7 Message
SSN Number	PID.19
Physician/CNM	PV1.17
Reason for Admission	PV2.3
Emergency Name	NK1.2
Emergency Contact Number	NK1.5
Relationship	NK1.3
Bed Number	PV1.3.3
Address	PID.11.1
City	PID.11.3
State	PID.11.4
Zip	PID.11.5

5.4 LND Update Screen:

Field	Field in HL7 Message
On Call MD	PV1.9
Nurse	PV1.52

6 Technical Support

PeriGen® IT Solutions Technical Support is available 24/7, 365 days a year. If you need help, please call: 1-855-236-4353.

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